

AI Full-Stack Developer

Candidate: Chirag N Sundar | **Role:** AI Full-Stack Developer

SUBMISSION WINDOW

Maximum 24 hours

Complete and submit as early as possible once you have a working solution.

A strong submission does not need to consume the full 24 hours.

About Quantum Weave

Quantum Weave is an AI-first business implementation venture of BrandMint AI Pvt. Ltd. We build AI-powered applications, AI employees, business automations, knowledge systems, SaaS products, API integrations and digital business systems that solve practical business problems.

An AI Full-Stack Developer at Quantum Weave is expected to think beyond coding and operate as an AI Solution Builder who can understand a business problem, design the system, build it, integrate intelligence and automation, and deploy a usable solution.

Business Scenario

A growing service business receives enquiries through its website and WhatsApp. The business currently faces the following problems:

- Leads are stored manually.
- Enquiries are not qualified consistently.
- Follow-ups are delayed.
- Business information is scattered across documents.
- Customer questions consume team time.
- Management cannot easily track leads and activities.

Your task is to build a functional prototype of an AI-Powered Lead & Customer Intelligence System.

The solution should connect lead management, AI intelligence, business knowledge, automation and customer interaction into one usable application.

Assignment Tasks

1. Full-Stack Application

Build a responsive application containing both a public side and an internal side.

- Public side: simple business/service page and lead/contact enquiry form.
- Internal side: login/authentication, lead dashboard, lead detail view, create/update lead, lead status/pipeline management and basic analytics.
- Suggested stages: New → Contacted → Qualified → Proposal → Won / Lost.
- Use a suitable modern frontend, backend and database architecture.

2. Backend, APIs & Database

Build the required backend and data layer.

- Support lead creation, retrieval, updates, stage changes, AI-generated information and activity/follow-up information.
- Expose suitable APIs and structure the data so the application could reasonably grow beyond the prototype.

3. AI Lead Intelligence

Use an LLM to analyse each enquiry and integrate the output into the lead workflow.

- Generate a lead summary, customer problem/requirement, intent, priority, suggested service, recommended next action and suggested follow-up response.
- The AI capability should be part of the business workflow rather than a standalone chatbot.

4. AI Knowledge Assistant / RAG

Create a simple business knowledge base using sample business/service documents or content.

- Retrieve relevant information from the knowledge source.
- Answer business/service questions using retrieved context.
- Handle questions where reliable information is unavailable.
- Use an appropriate RAG/vector-search approach.

5. AI Sales / Support Agent

Create at least one AI agent that can work with business context.

- The agent should understand the customer enquiry, retrieve business knowledge, recommend an appropriate service, suggest the next action and generate a contextual response.
- Demonstrate at least one example of function/tool calling, structured AI action, agent-to-application interaction or API invocation.
- The objective is to show that AI can participate in a workflow rather than only generate text.

6. Business Automation

Implement one meaningful automated workflow.

- Example: New enquiry → Lead created → AI analyses lead → Priority assigned → Follow-up generated → Database/CRM updated → Team notification or follow-up action triggered.
- Use a suitable workflow approach such as n8n, Make, Zapier, custom APIs/webhooks or another appropriate method.

7. WhatsApp / External Integration

Demonstrate one external business integration.

- Preference may be given to a WhatsApp-style conversational workflow, but another useful integration may be selected where appropriate.
- Possible examples: WhatsApp Cloud API, email, Google Calendar/Google APIs, CRM, Meta APIs or another relevant service.
- If live credentials or API access are unavailable, create a technically meaningful API/webhook simulation and explain how the live integration would work.

8. Reliability & Security

Handle important failure and security cases.

- Consider invalid input, missing information, AI/API failure, duplicate leads, authentication failure and unexpected LLM responses.
- Do not expose API keys, secrets, passwords or credentials.
- AI-generated outputs should remain reviewable by a human where appropriate.

9. Testing

Demonstrate testing for the most important parts of the system.

- Choose the areas you believe deserve testing most, such as backend APIs, important business logic, authentication, AI output handling or the automation workflow.

10. Deployment

Deploy the working prototype.

- Use any suitable cloud/deployment stack.
- The application should be accessible for demonstration wherever reasonably possible.

11. Technical Documentation

Provide concise documentation covering the solution.

- Include: problem being solved, architecture, technology choices, frontend/backend/database, AI implementation, RAG approach, agent/tool workflow, automation, external integration, testing, security considerations, deployment, known limitations and production improvements.
- Include a simple architecture diagram.

Expected Outcome

The solution should demonstrate an end-to-end flow similar to:

Customer enquiry → Lead captured → Database → AI analysis → Knowledge retrieval → AI recommendation/action → Pipeline update → Automated follow-up/action → Dashboard visibility → Deployed application

The goal is not to build every possible feature. The goal is to demonstrate that you can connect **Full-Stack Development + AI + Knowledge + Agents + Automation + APIs + Deployment** into one practical business solution.

Timing

Maximum submission window: 24 hours from receiving the assignment.

You are encouraged to complete and submit the assignment as early as possible once you have a working solution.

A strong submission does not need to consume the full 24 hours.

Prioritise in this order:

1. Working end-to-end flow
2. Important functionality
3. Reliability
4. Documentation
5. Additional enhancements

Do not spend excessive time polishing low-impact UI details while important functionality remains incomplete.

If a complete production-grade implementation is not practical within the available time, build the most important functionality and clearly explain what you completed, what remains, how you would implement the remaining parts, and what you would improve with additional time.

Candidate Readiness

You should understand your submitted solution and be prepared to explain your architecture, code, AI workflows, technical decisions, limitations and production improvements. You may also be asked to modify or troubleshoot part of the solution during the next stage.

Wherever reasonably possible, do not submit only documentation or strategy. Build and demonstrate the solution.